

PRACTICAL COURSE WORKBOOK

Open Data Analysis for Policy

Use evidence to explain a public-interest problem

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	an evidence brief with a reproducible analysis
SKILLS	Open data, Python, Pandas, World Bank, Policy analysis, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use Open data appropriately in a realistic, bounded task.
- Use Python appropriately in a realistic, bounded task.
- Use Pandas appropriately in a realistic, bounded task.
- Use World Bank appropriately in a realistic, bounded task.

WHO THIS IS FOR

Public-interest learners using evidence and technology responsibly.

BEFORE YOU BEGIN

- Basic research and spreadsheet skills

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

Open Data Analysis for Policy is a structured, practical course covering Open data, Python, Pandas, World Bank, Policy analysis. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Open data: Public problems	1. Stakeholders with Open data 2. Institutions with Python 3. Evidence with Pandas
02 Python: Digital methods	1. Open data with World Bank 2. Services with Policy analysis 3. Participation with Open data
03 Pandas: Policy design	1. Options with Python 2. Impact with Pandas 3. Implementation with World Bank
04 World Bank: Accountability	1. Measurement with Policy analysis 2. Transparency with Open data 3. Maintenance with Python

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Open Data Analysis for Policy project using Open data, Python, Pandas.

DELIVERABLE	A working Open Data Analysis for Policy artefact plus an evidence-based self-review.
TOOLS	A suitable Open data environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Open data.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Open Government Data Toolkit

World Bank - reviewed 2026-08-21

<https://opendatatoolkit.worldbank.org/en/data/opendatatoolkit/home>

Digital government

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/digital-government.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Open data scope.